

Name:

TAKE-HOME QUIZ (BONUS)

TRANSPORT PHENOMENA I ChE 311

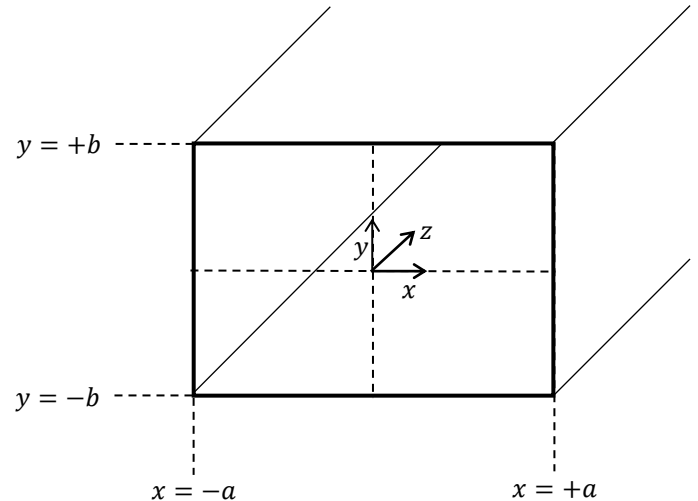
A fluid with constant density ρ and viscosity μ flows through a rectangular duct of length L . The modified pressure at the inlet and the outlet of the duct are $\mathcal{P}_0$ and $\mathcal{P}_L$.

Partial differential equation for the steady-state velocity profile v_z as a function of x and y .

$$\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} = \frac{\Delta \mathcal{P}}{\mu L}$$

The solution to the above PDE as a double summation over the product of orthogonal sine and/or cosine functions is:

$$v_z = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2m-1}{2b}\pi y\right)$$



- 1- Calculate C_{nm} 's by substituting the above double summation in the Navier-Stokes PDE and using cosine orthogonality relations.
- 2- Plot dimensionless velocity $\frac{v_z}{\left(\frac{-\Delta \mathcal{P} a^2}{\mu L \pi^4}\right)}$ as a function of $\frac{x}{a}$ and $\frac{y}{b}$ assuming the aspect ratio of the duct $\frac{a}{b} = 1$.

$$\begin{aligned} \frac{\partial^2}{\partial x^2} \left[\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \cdot \cos\left(\frac{2n-1}{2a}\pi x\right) \cdot \cos\left(\frac{2m-1}{2b}\pi y\right) \right] + \frac{\partial^2}{\partial y^2} \left[\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \cdot \cos\left(\frac{2n-1}{2a}\pi x\right) \cdot \cos\left(\frac{2m-1}{2b}\pi y\right) \right] &= \frac{\Delta \mathcal{P}}{\mu L} \\ \frac{\partial^2}{\partial x^2} \left[\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \cdot \cos\left(\frac{2n-1}{2a}\pi x\right) \cdot \cos\left(\frac{2m-1}{2b}\pi y\right) \right] + \frac{\partial^2}{\partial y^2} \left[\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \cdot \cos\left(\frac{2n-1}{2a}\pi x\right) \cdot \cos\left(\frac{2m-1}{2b}\pi y\right) \right] &= \frac{\Delta \mathcal{P}}{\mu L} \\ C_n \frac{\partial^2}{\partial x^2} \left[\sum_{n=1}^{\infty} \cos\left(\frac{2n-1}{2a}\pi x\right) \right] &= C_n \frac{\partial^2}{\partial y^2} \left[\sum_{m=1}^{\infty} \cos\left(\frac{2m-1}{2b}\pi y\right) \right] \\ \frac{(2n-1)\pi}{2a} \cdot -\sin\left(\frac{2n-1}{2a}\pi x\right) &= C_m \frac{2m-1}{2b} \pi \cdot -\sin\left(\frac{2m-1}{2b}\pi y\right) \\ C \sum_{n=1}^{\infty} \left(\left(\frac{(2n-1)\pi}{2a} \right)^2 \cdot -\cos\left(\frac{2n-1}{2a}\pi x\right) \right) &= C \left(\left(\frac{2m-1}{2b} \pi \right)^2 \cdot -\cos\left(\frac{2m-1}{2b}\pi y\right) \right) = \frac{\Delta \mathcal{P}}{\mu L} \\ \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \left(-\left(\frac{(2n-1)\pi}{2a} \right)^2 - \left(\frac{(2m-1)\pi}{2b} \right)^2 \right) \cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2m-1}{2b}\pi y\right) &= \frac{\Delta \mathcal{P}}{\mu L} \end{aligned}$$

$$\cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2m-1}{2b}\pi y\right) \cdot \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \left[-\left(\frac{2n-1}{2a}\pi\right)^2 - \left(\frac{2m-1}{2b}\pi\right)^2 \right] \cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2m-1}{2b}\pi y\right) = \frac{\Delta\phi}{\mu L} \cdot \cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2m-1}{2b}\pi y\right)$$

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \left[-\left(\frac{2n-1}{2a}\pi\right)^2 - \left(\frac{2m-1}{2b}\pi\right)^2 \right] \cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2n-1}{2a}\pi x\right) \cdot \cos\left(\frac{2m-1}{2b}\pi y\right) \cdot \cos\left(\frac{2m-1}{2b}\pi y\right) = \frac{\Delta\phi}{\mu L} \cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2m-1}{2b}\pi y\right)$$

$$\int_{-a}^a \cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2n'-1}{2a}\pi x\right) dx = \begin{cases} 0, & n \neq n' \\ a, & n = n' \end{cases} \Rightarrow a C_1$$

$$\int_{-b}^b \cos\left(\frac{2m-1}{2b}\pi y\right) \cdot \cos\left(\frac{2m'-1}{2b}\pi y\right) dy = \begin{cases} 0, & m \neq m' \\ b, & m = m' \end{cases} \Rightarrow b C_2$$

By orthogonality, all terms vanish except when $n=n'$ and $m=m'$

$$C_{nm} \left[\left(\frac{2n-1}{2a}\pi\right)^2 + \left(\frac{2m-1}{2b}\pi\right)^2 \right] (ab) = -\frac{\Delta\phi}{\mu L} \int_{-b}^b \cos\left(\frac{2m-1}{2b}\pi y\right) \cdot \cos\left(\frac{2m-1}{2b}\pi y\right) dy \int_{-a}^a \cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2n-1}{2a}\pi x\right) dx$$

$$C_{nm} \left[\left(\frac{2n-1}{2a}\pi\right)^2 + \left(\frac{2m-1}{2b}\pi\right)^2 \right] (ab) = -\frac{\Delta\phi}{\mu L} \cdot \frac{4a}{(2n-1)\pi} \cdot \frac{4b}{(2m-1)\pi}$$

$$\frac{C_{nm} \left[\left(\frac{2n-1}{2a}\pi\right)^2 + \left(\frac{2m-1}{2b}\pi\right)^2 \right] (ab)}{\left[\left(\frac{2n-1}{2a}\pi\right)^2 + \left(\frac{2m-1}{2b}\pi\right)^2 \right]} = \frac{-\frac{\Delta\phi}{\mu L} \frac{16ab}{(2n-1)(2m-1)\pi^2}}{\left[\left(\frac{2n-1}{2a}\pi\right)^2 + \left(\frac{2m-1}{2b}\pi\right)^2 \right]}$$

$$C_{nm} = -\frac{\Delta\phi}{\mu L} \frac{16}{(2n-1)(2m-1)\pi^2} \cdot \frac{1}{\left(\frac{2n-1}{2a}\pi\right)^2 + \left(\frac{2m-1}{2b}\pi\right)^2}$$

$$V_z = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} -\frac{\Delta\phi}{\mu L} \frac{16}{(2n-1)(2m-1)\pi^2} \cdot \frac{1}{\left(\frac{2n-1}{2a}\pi\right)^2 + \left(\frac{2m-1}{2b}\pi\right)^2} \cdot \cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2m-1}{2b}\pi y\right)$$

$$V = \frac{V_z}{-\frac{\Delta\phi a^2}{\mu L \pi^4}} = \frac{-\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} -\frac{\Delta\phi}{\mu L} \frac{16}{(2n-1)(2m-1)\pi^2} \cdot \frac{1}{\left(\frac{2n-1}{2a}\pi\right)^2 + \left(\frac{2m-1}{2b}\pi\right)^2} \cdot \cos\left(\frac{2n-1}{2a}\pi x\right) \cos\left(\frac{2m-1}{2b}\pi y\right)}{-\frac{\Delta\phi a^2}{\mu L \pi^4}}$$

$$V = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{16}{(2n-1)(2m-1)} \frac{\cos\left(\frac{2n-1}{2}\pi x^*\right) \cos\left(\frac{2m-1}{2}\pi y^*\right)}{(2n-1)^2 + (2m-1)^2}$$

$$\begin{aligned} x^* &= \frac{x}{a} \\ y^* &= \frac{y}{b} \end{aligned} \quad \frac{a}{b} = 1$$

